Waiting for the first success

Upper high school · probability and sequences · Lesson and method · 08

Prerequisites. Independent trials, geometric sequences and sums.

Key point · Ideas and method. Repeat independent trials, each with success probability p, where $0 < p \leq 1$. T is the trial number of the first success, starting at 1.
The event $T = k$ requires $k - 1$ failures followed by success: $P(T = k) = (1 - p)^{k-1}p$. Likewise, $P(T > n) = (1 - p)^n$ for $n \geq 0$.
The expectation is $E(T) = \frac{1}{p}$. It follows from the tail-sum formula $E(T) = \sum P(T > n)$, for $n \geq 0$, and the geometric series. This is an average, not a guaranteed success time.

Example 1. For a fair die with success defined as rolling 6, $p = \frac{1}{6}$. The mean waiting time is 6 rolls, but the probability of waiting more than 6 rolls is $(\frac{5}{6})^6$, about 0.335.

Activity 1 · Explore. For $p = \frac{1}{4}$, compare $P(T > n)$ for $n = 0, 1, \ldots, 10$. Relate the decrease to the geometric factor $\frac{3}{4}$.